DAYS 36 TO 40 | CAPSTONE

Capstone Guide

Five days, one real-world product, built end to end by your team.

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1. What the capstone is

For the final week you work in teams of 3 to 4 and build a complete product for a realistic customer scenario, using everything from Days 1 to 35: Python, APIs, databases, React, testing, data pipelines, cloud and containers, GenAI, agents, security, evaluation and LLMOps. You also practise the FDE skills: scoping with a customer, handling ambiguity, communicating progress and presenting results.

Goal: a working, tested, deployable product with measured quality, and a confident ten minute demo. Depth beats breadth: a smaller scope done well scores higher than many half-finished features.

2. Choose one of five projects

#	Project	Core skills
01	ShopSmart: E-commerce Recommendation and Shopping Assistant	Embeddings for product descriptions; hybrid search with RRF
02	BookFinder: Library Book Finder with Natural Language Search	Embeddings and hybrid search with RRF
03	PolicyPal: Enterprise HR and IT Policy Chatbot	Context engineering and chunking
04	SupportPilot: Agentic Customer Support Copilot	Tool calling and LangGraph
05	InsightDesk: Retail Sales Pipeline and Natural Language Data Agent	Data agent with sandboxed code

Full briefs are in the Projects folder. Each team picks one project in the first hour of Day 36. Two teams may choose the same project; they will be compared on the same checklist.

3. Technical baseline (every project)

- FastAPI back end with Pydantic models and at least 6 endpoints (Days 5 to 8)
- A database: SQL (SQLAlchemy) and/or MongoDB (Days 4, 6 to 9)
- React front end with loading, error and empty states (Days 11, 12)

- An AI component from Weeks 5 and 6: RAG, recommendation with embeddings, agent or data agent (Days 21 to 27)
- Tests with PyTest and a GitHub or Azure DevOps CI pipeline (Days 13, 17, 18)
- Dockerfile and docker-compose; Azure deployment or documented steps (Days 18, 19, 33)
- Guardrails and secrets handling (Days 8, 28)
- Golden set, metrics and a quality gate (Days 29, 30)
- Structured logs and a monitoring report or dashboard (Days 34, 35)

4. Day-by-day plan

Day 36 (Mon Dec 14): Kick-off and design

- Quiz on Day 35
- Teams of 3 to 4 formed; project chosen
- Problem statement, users, success metrics, assumptions log
- Architecture diagram and data model
- Repository, board and CI skeleton created
- Checkpoint 1 (end of day): design review with mentor

Day 37 (Tue Dec 15): Core build

- Stand-up (15 min)
- Data loaded; back end API with main endpoints
- AI core working end to end (retrieval, recommendation, agent or pipeline)
- Golden set draft of 20 items
- Checkpoint 2: walking skeleton demo

Day 38 (Wed Dec 16): Complete features and quality

- Stand-up
- Front end connected to the API
- Guardrails and error handling
- Evaluation run and first error analysis
- Tests in CI; quality gate added
- Checkpoint 3: feature-complete MVP

Day 39 (Thu Dec 17): Harden, deploy, observe

- Stand-up
- Docker images; deploy to Azure or a documented local deployment
- Structured logs and a monitoring view
- Red-team and slice checks; fix top issues
- Demo rehearsal with peer feedback
- Code freeze at 17:00 IST

Day 40 (Fri Dec 18): Demo day

- Final rehearsal (morning)

- Demo: 10 minutes + 5 minutes Q&A; per team
- Evaluator scoring with the rubric
- Individual reflection and peer feedback
- Closing and certificates

5. Daily stand-up (15 minutes)

Each person answers three questions: what I finished yesterday, what I will finish today, what is blocking me. The team updates its board and decision log. Mentors join at least once a day.

6. Rules

- AI coding assistants are allowed and encouraged. Disclose in the README which parts were generated and how you reviewed them.
- Use only the generated or public sample data. Never use real customer or personal data.
- Keep secrets out of Git. Use environment variables or a secret store.
- Every team member must be able to explain every part of the system during Q&A.;
- Cloud resources: use tags, stay within the agreed budget, and delete resource groups after demo day.

7. Deliverables (due Day 40, 09:00 IST)

Deliverable	Details
Repository	Code, tests, Dockerfiles, pipeline YAML, golden set, README with setup, architecture and AI tool disclosure
Architecture diagram	Components, data flow, where AI is used, guardrails
Evaluation report	Golden set results, metrics, error analysis, slices, known limits
Operations note	How to deploy, monitor, roll back; one runbook
Demo deck	8 to 12 slides: problem, solution, demo, results, limits, next steps
Decision log	Key decisions with options, reasons and dates

8. Demo day format

- 10 minutes: problem and users (1), live demo (5), results and evaluation (2), limits and next steps (2).
- 5 minutes of Q&A; from evaluators acting as the customer's sponsor, IT lead and security lead.
- Every team member speaks. Lead with the answer, show evidence, be honest about limits.

9. Evaluation rubric (100 points)

Criterion	Points	What evaluators look for
Problem understanding and scoping	10	Clear problem statement, users, success metrics, assumptions log, agreed MVP scope (Days 15, 20).
Architecture and design	10	Sensible architecture diagram, layered code, right tool choices with reasons, data model (Days 4, 6, 7, 22, 33).
Core functionality	20	All must-have features work end to end in the demo; project checklist items delivered.
AI and data quality	15	Golden set of at least 20 items, measured metrics (retrieval, faithfulness or recommendation quality), error analysis (Days 23, 29, 30).
Engineering quality	10	Readable code, tests (API, pipeline, prompts), Git history with pull requests, CI pipeline green (Days 13, 17, 18).
Security, guardrails and responsible AI	10	Input and output guardrails, secrets handling, least privilege, a red-team check, fairness or language slice check (Days 8, 28).
Deployment and operations	10	Containerised, deployable (Azure or documented), structured logs, a dashboard or report, one drift or quality check (Days 19, 31 to 35).
Demo and communication	10	Answer-first readout, live demo, honest limits, confident Q&A;, clear handover README (Days 15, 20, 24).
Teamwork and process	5	Daily stand-ups, decision log, fair contribution, kind and specific feedback (Days 15, 24, 31).

Each criterion is scored on four levels; points = weight x level / 4.

Level	Meaning
4 Excellent	Goes beyond the brief; production-minded; evidence for every claim.
3 Good	Meets the brief fully with minor gaps.
2 Developing	Partly meets the brief; important gaps or untested parts.
1 Beginning	Major parts missing or not working.

Grade bands: 85 to 100 Distinction, 70 to 84 Merit, 55 to 69 Pass, below 55 Needs resubmission. Individual scores can be adjusted by up to 10 points based on peer feedback and Q&A.;

10. Submission checklist

- README lets a stranger run the project in under 15 minutes
- CI pipeline is green on main
- Golden set and evaluation report are committed
- No secrets or personal data in the repository
- Demo works without internet surprises (have a recorded backup)
- Azure resources tagged and scheduled for deletion

Ganbatte (頑張って): do your best